

When long-term UK borrowing costs rise And Bank Rate stays put

A short guide to the UK long-rate model

`uk-long-rate/` in MacroModellerBot

September 2026

The question

What happens when long-term UK borrowing costs rise, while Bank Rate stays put?

- Long-term rates are what mortgages, firms, and the government pay to borrow for years, not the overnight policy rate.
- They can rise because investors demand extra compensation for holding long debt, even if the Bank does not change Bank Rate.
- This talk looks at that case: a one-off rise of 100 basis points in long rates, with Bank Rate held on its baseline path for three years.
- Every result below is that experiment, measured against a baseline in which the rise does not happen.

In the model

- **Mortgages.** Most deals are fixed. The higher rate bites as those deals roll off, not all at once.
- **House prices.** Dearer new mortgages pull house prices down, which weakens borrowers who rely on the house as collateral.
- **Business investment.** A higher long-term cost of capital leads firms to invest less.
- **The government interest bill.** New and refinanced gilts cost more. Part of the stock is index-linked, so the bill does not jump immediately.
- **Sterling.** With Bank Rate held fixed, the pound strengthens in this experiment.

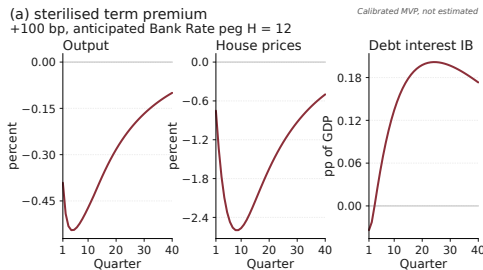
Described, but not modelled

- Pension-fund (LDI) stress, as in September 2022.
- A sovereign-risk shock, in which gilt worries also hit the pound.

Why timing matters

- Only about 9% of the mortgage stock reprices each quarter. That share is $\lambda_q = 0.09$.
- A borrower whose fix has years left does not pay the new rate yet. Someone remortgaging this quarter does.
- So the average mortgage rate on the whole stock climbs slowly, even when the rate on new loans jumps.
- The pressure on household spending therefore builds over several quarters, rather than arriving in one month.

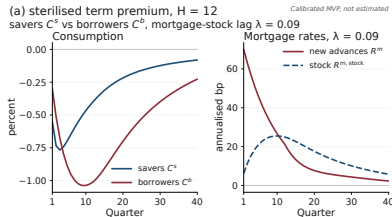
What the model says



Response to a one-off +100 bp rise in long rates, measured against a baseline without it.

- Output is 0.55% lower than it would otherwise be (trough at quarter 4).
- House prices are 2.6% lower than otherwise (quarter 8).
- The government interest bill is 0.20 pp of GDP higher than otherwise (peak at quarter 24).

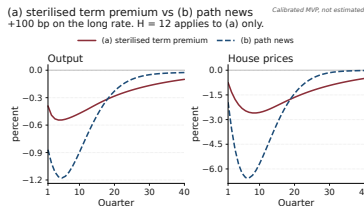
Who is hit



Same one-off +100 bp rise in long rates, against a baseline without it.

- Savers cut back first: 0.77% below baseline at quarter 3.
- Borrowers cut back more, and later: 1.04% below baseline at quarter 9, as their mortgages reprice.
- That lag is the 9% quarterly reset. The rate on new loans jumps immediately; the rate on the whole stock does not.

If markets instead expect Bank Rate to rise



Both lines are a one-off +100 bp rise in long rates, against a baseline without it. Only the solid line keeps Bank Rate on hold.

- If the long rate rises because Bank Rate itself is expected to rise, Bank Rate is about 315 bp higher than otherwise on impact, and about 560 bp higher at quarter 4.
- Output is then 1.178% lower than otherwise (quarter 5), and house prices 6.551% lower (quarter 7).
- That is a deeper hit than when Bank Rate stays put (0.55% and 2.6%).

- The numbers come from a calibrated model. They are not estimates fitted to UK data.
- A few settings are provisional choices, including how strongly inflation responds and how house prices are determined.
- The sovereign-risk scenario is not yet modelled.
- LDI stress is not modelled. The September 2022 gilt spiral is a separate, non-linear event.

- Estimate the effects directly from high-frequency surprises in gilt yields, using local projections.
- Include a state-dependent version, so the effect can differ when the gilt market is stressed.
- Then estimate this model on UK data, with a wide prior on how strongly inflation responds.
- The sovereign-risk scenario is still to be built.

Appendix: three fixed settings

- About 9% of mortgages reprice each quarter ($\lambda_q = 0.09$).
- A quarter of the gilt stock is index-linked (share 0.25).
- Average gilt duration is 9.1 years, so the government's coupon book refixes slowly.

The rate on the whole mortgage stock is mostly last quarter's rate:

$$R_t^{\text{stock}} = (1 - \lambda_q) R_{t-1}^{\text{stock}} + \lambda_q R_t^{\text{new}}.$$

With $\lambda_q = 0.09$, nine tenths of the stock keeps the old rate each quarter.